

Syllabus

October 18, 2025

Course

Quantum Field Theory I, 2025 fall

Course Meeting Times

Lectures: 2 sessions / week, 1.5 hours / session

Prerequisite

You must complete *Quantum Mechanics* before attempting this course.

Description

Quantum Field Theory I is a one-term self-contained subject in quantum field theory. Concepts and basic techniques are developed through applications in elementary particle physics and condensed matter physics.

Required Textbooks

- Peskin, Michael E., and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. CRC Press, 1995. ISBN: 9780201503975.
- Weinberg, Steven. *The Quantum Theory of Fields*, Volume 1: Foundations. Cambridge University Press, 2005. ISBN: 9780521670531.

Topics

- Why quantum field theory
 - Principle of locality: classical field theories
 - Action principle, Lagrangian and Hamiltonian
 - Symmetry and Noethers theorem
 - Implications of relativistic symmetry
 - What is wrong with relativistic quantum mechanics
 - Special relativity plus quantum mechanics requires quantum field theory
 - Continuum limit of discrete systems: Many condensed matter applications
- Free scalar field theories:
 - Canonical quantization of a free scalar field
 - Particle interpretation
 - Propagators
 - Complex scalar fields
- Interactions: path integral approach

- Path integrals for quantum mechanics
- Path integral for quantum scalar fields
- Perturbation theory: Feynman diagrams
- Cross section and scattering matrix
- Dirac theory
 - Dirac equation and its Lorentz covariance
 - Canonical quantization
 - Spin and statistics
 - Discrete symmetries
 - Path integrals for Dirac fields
- Maxwell theory
 - Gauge symmetry
 - Canonical quantization
 - Path integral quantization
- Quantum electrodynamics (may cover depending on time)
 - Feynman rules
 - Elementary processes
 - Compton and inverse Compton scatterings

Grading

There is **no exam** for this course. The course grade will be based 60% on homework, 30% on class participation and 10% on class performance.

Homework

Homework is normally posted on the course website on Monday and will be due on Monday one week later. Any late homework will be only counted as half credit. For example, if you get 90% on a late homework, it will only count as 45%. However, your lowest homework score will be discarded at the end of the semester; only the remaining $n - 1$ will be used in determining your grade.

!!! Homework is a very important part of this course. I believe that sitting down yourself and trying to go through a problem not only helps you learn the material deeply, but also develops analytical tools to a successful career in science. I understand the difficulties in working on the homework in QFT, therefore I encourage each student to make his/her own attempt on every problem, no matter using AI, searching online or working together with your classmates. However, the solution you submit must reflect on your own work. They must not copy and paste other's work. **Plagiarism is a serious offense** and is easy to recognize. Don't submit work which is not your own. It's quite easy for me to figure out the plagiarism. And once I figure out your homework is a plagiarism, your grade for this homework will be 0.